



Grade 11 Geography

8. 🌐 Mass Movements Impacts & Solutions

Learning Guide

1. Impacts of Mass Movements

Mass movements can affect both **people** and the **environment**. Their effects may be severe, especially when they occur near settlements, farms, roads, and rivers.

2. Impact on People

Loss of Life and Injuries

- People may be killed or injured when:
 - Rocks fall.
 - Slopes collapse.
 - Landslides or mudflows move through settlements.
- Rapid mass movements are especially dangerous because they may happen with little warning.

Homes and Possessions Destroyed

- Houses can be:
 - Buried.
 - Damaged.
 - Completely destroyed.
- People may lose personal belongings and may be forced to leave their homes.

Farms and Livestock Damaged

- Farmland may be covered or washed away.
- Crops may be destroyed.
- Livestock may be injured or killed.

Buildings Collapse

- Movement of the ground may weaken the foundations of buildings.
- Buildings may crack, tilt, or collapse.

Transport Networks and Infrastructure Damaged

- Roads, bridges, and other infrastructure can be damaged.
- Debris may block roads and make transport difficult.
- The photographs on page 3 show examples of **damaged roads, vehicles affected by debris, damaged farmland, and buildings near unstable slopes.**

High Repair Costs

- Governments, businesses, and individuals may have to spend large amounts of money repairing:
 - Roads.
 - Buildings.
 - Farms.
 - Infrastructure.

Key idea: Mass movements can cause loss of life, destroy property, damage infrastructure, and create major financial losses.

3. Impact on the Environment

Mass movements also change the natural environment.

Loss of Topsoil and Vegetation

- Mass movements can remove:
 - **Topsoil**
 - **Vegetation**
 - Even entire **forests**
- Once the fertile topsoil is removed, it becomes more difficult for plants to grow again.

Exposure of Bedrock

- When soil is removed, **bedrock may be exposed**.
- Bedrock provides poor conditions for new plants to establish.
- This slows the recovery of vegetation.

 **Key idea:** *Loss of soil makes revegetation more difficult.*

Blocking of Rivers

- Landslides can fall into river channels.
- The material may **block the river**.
- This may create a natural **dam**.

Sediment Pollution in Rivers

- Loose soil and rock may enter rivers.
- This increases the amount of **sediment** in the water.
- The document links this to **pollution**, which may harm fish.

 **Sequence:** *Mass movement → sediment enters river → pollution → fish harmed*

Destruction of Animal Habitats

- Animals may lose their habitats when:
 - Vegetation is destroyed.
 - Forests are removed.
 - Slopes collapse.
 - Rivers are altered.

4. How Mass Movements Reshape the Landscape

Mass movements are also part of the natural process that changes landscapes over time.

Mountains Shrink

- Material is removed from higher slopes.
- Over time, mountains may become lower.

Valleys Fill

- Material transported downslope may collect in valleys.
- This gradually fills low-lying areas.

Slopes Recede

- Material is removed from the slope face.
- The slope gradually moves backwards.

 **Key idea:** Mass movement removes material from high areas and deposits it in lower areas, gradually reshaping the landscape.

5. Development of Rills, Gullies and Valleys

The visual on page 5 shows the sequence:

Rills → Gullies → Valley

- Small channels may develop into larger erosion features.
- Over time, these features become wider and deeper.
- This contributes to the development and reshaping of valleys.

6. Strategies to Prevent or Minimise Mass Movements

Mass movements cannot always be completely prevented, but their effects can be **reduced**.

1. Keep Vegetation on Slopes

- Plant roots help **hold soil together**.
- Roots stabilise the slope.
- Vegetation therefore reduces the chance of soil moving downslope.

 **Key idea:** *More vegetation = stronger and more stable slope.*



2. Avoid Building Large Structures on Steep Slopes

- Large buildings add **extra weight** to slopes.
- This can make unstable slopes more likely to fail.
- Development should therefore be avoided on very steep slopes.



3. Keep Mine Dumps from Becoming Too Steep

- Mine dumps can become unstable if their sides are too steep.
- Making the slopes gentler reduces the chance of collapse.



4. Build Drainage Channels

- Drainage channels remove **excess water** from slopes.
- Too much water can:
 - Add weight.
 - Saturate soil.
 - Reduce slope stability.
- Removing excess water helps keep the slope more stable.



Key idea: Good drainage reduces water-related slope failure.



5. Check Rock Structure Before Development

- Rock structure should be examined before constructing on slopes.
- Weaknesses in the rock may increase the risk of movement.
- Development should therefore consider the stability of the underlying rock.



7. Engineering Structures

Engineering structures can be used to stabilise slopes and protect roads, buildings, and people.

Gabions

- Used to strengthen and stabilise slopes.
- Help reduce the movement of loose material.

Retaining Walls

- Built to support unstable slopes.
- Help hold soil and rock in place.

Rock Nets

- Used particularly on:
 - **Steep slopes**
 - **Road cuttings**
- Catch falling rocks before they reach roads or other structures.



Key idea: Engineering structures help hold material in place or prevent falling debris from causing damage.



8. Impact and Prevention Link

Learners should understand how each strategy deals with a particular problem:

Problem	Strategy
Loose soil	Keep vegetation
Too much water	Build drainage channels
Heavy buildings on unstable slopes	Avoid construction on steep slopes
Unstable mine dumps	Reduce slope steepness
Weak rock structure	Check rock before development
Falling rocks	Rock nets
Unstable slopes	Retaining walls / gabions



9. Final Exam Summary

- Mass movements affect both **people and the environment**.
- Effects on people include:
 - Loss of life.
 - Injuries.

- Destruction of homes.
 - Damage to farms and livestock.
 - Building collapse.
 - Damage to roads and infrastructure.
 - High repair costs.
 - Environmental impacts include:
 - Loss of topsoil.
 - Loss of vegetation and forests.
 - Exposure of bedrock.
 - Blocked rivers.
 - Sediment pollution.
 - Destruction of animal habitats.
 - Changes to mountains, valleys, and slopes.
 - Risk can be reduced by:
 - Keeping vegetation.
 - Avoiding large buildings on steep slopes.
 - Keeping mine dumps less steep.
 - Building drainage channels.
 - Checking rock structure.
 - Using gabions.
 - Building retaining walls.
 - Installing rock nets.
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